Synergy '26 / StackSense Challenge

StackSense

Intelligent Tech Stack Recommendation Engine

A deterministic architecture workbench for evaluating project constraints.

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Public repository copy. Entrant email appears in the submission copy.

github.com/shinjan-2006/StackSense

10 November 2026

1. Abstract

StackSense converts a structured project scenario into an architecture recommendation, cost band, trade-offs and warnings. It runs offline in a browser or through a dependency-free Node CLI. The engine validates the shared schema, selects the deployment boundary and scores compatible backend options. Supporting services and cost are evaluated alongside the core stack. Generated JSON preserves the drivers, alternatives and planning assumptions used in the decision.

2. Problem understanding

The useful result is a stack a team can investigate within its constraints. The engine rejects explicit deployment conflicts and estimates budget fit before returning a stack. Delivery speed, language fit, workload and operating burden guide selection. Unknown regions and external data paths are surfaced for review. Provider products are design candidates, not compliance certifications.

3. System architecture

```
Browser form / JSON file / HTTP request
    -> shared schema and validation
    -> constraints, region and provider
    -> backend scores and supporting layers
    -> capacity, cost and budget gates
    -> recommendation JSON and browser results
```

schema.js defines field types and ranges. engine.js applies defaults, validates, and generates the same result in both runtimes. app.js maps form controls to the JSON scenario and presents cost, conflicts and component details. cli.js supports individual and batch exports. The optional HTTP server provides the same engine over a local endpoint.

4. Input design and research

Project type and requested platforms drive architecture shape and client coverage. Name identifies the project and description provides context. User counts, concurrency, traffic pattern, workload and volume inform scale warnings, backend scoring, caching and storage. Team size, languages, experience and DevOps capacity inform framework fit and operational warnings. Budget and delivery horizon set the cost gate and delivery weighting.

Cloud, deployment model, residency and existing systems resolve the provider and region. Requirements cover identity, realtime, offline behavior, search, storage, payments, tenancy, ML, SEO and service targets. Compliance flags require control and contract review. Priorities steer scoring, search and cache selection, and resilience allowances.

Optional inputs cover throughput, recovery, sensitivity, portability, search size, concurrent editing, ledger model, offline writes and resumable uploads. They change sizing, safeguards or selected components. The shared schema rejects malformed sections, unknown fields and values, missing required inputs and invalid numeric ranges.

The browser uses HTML5, CSS and JavaScript. The CLI and optional HTTP server use Node.js 18 or later and were tested on Node.js 24. Both entry points use the same dependency-free engine.

4. Input design and research (continued)

scenario_id · Optional · string. labels JSON and batch filenames

compliance · Optional · unique values from gdpr, dpdp, hipaa, pci_dss, rbi_localisation, soc2, iso27001. adds applicable scope checks and can restrict production

project.name · Optional · string. labels the project in the form

project.description · Optional · string. can infer ledger needs in fintech

project.type · Required · one of content_site, web_app, saas_b2b, ecommerce, marketplace, social, realtime_collab, fintech, healthtech, data_analytics, iot, ai_ml_product, internal_tool, gaming, api_platform, other. selects architecture and can infer live editing

project.platforms · Required · unique values from web, android, ios, desktop, api_only. selects clients and rejects API-only combinations

scale.users_at_launch · Required · safe integer ≥ 0. records launch context

scale.users_12_months · Required · safe integer ≥ 0. drives zero-budget scale checks and pilot scope

scale.peak_concurrent_users · Optional · safe integer ≥ 0. estimates request load and informs scoring or cache rules

scale.data_volume_gb_12_months · Optional · number 0–9007199254740991. sizes storage allowances

scale.traffic_pattern · Optional · one of steady, spiky, seasonal. informs cache selection

scale.workload · Optional · one of read_heavy, balanced, write_heavy. informs cache and relational storage reasoning

team.size · Required · safe integer ≥ 1. weights skill fit and flags solo operations risk

team.experience · Required · one of beginner, intermediate, senior, mixed. weights skill fit and flags beginner operations risk

team.languages · Required · list of non-empty strings. scores backend familiarity

team.has_devops · Optional · true/false. informs operations and latency warnings

budget.monthly_infra_usd · Required · number ≥ 0. gates the stack or yields a limited pilot

timeline.mvp_weeks · Required · safe integer ≥ 1. weights delivery and checks platform scope

infrastructure.cloud_preference · Optional · one of any, aws, gcp, azure, none. selects the provider ahead of ecosystem hints

infrastructure.deployment_model · Optional · one of managed_cloud, self_hosted_cloud_vm, on_premise, air_gapped. sets managed, VM or local hosting

infrastructure.data_residency · Optional · one of none, india, eu, us, other. restricts data placement

infrastructure.existing_systems · Optional · list of non-empty strings. informs provider fit, integration and lifecycle warnings

requirements.realtime · Optional · true/false. adds live transport

requirements.offline_support · Optional · true/false. adds local persistence and sync

requirements.full_text_search · Optional · true/false. selects an index

requirements.file_media_storage · Optional · true/false. adds upload storage

requirements.payments · Optional · true/false. adds hosted checkout within the allowed boundary

requirements.multi_tenant · Optional · true/false. keeps the workload on an application stack

requirements.seo_critical · Optional · true/false. favors pre-rendered or server-rendered web output

requirements.auth · Optional · one of none, email_password, social, phone_otp, enterprise_sso. selects identity and checks external dependencies

requirements.ml_features · Optional · one of none, inference_api, llm_rag, training. selects an inference, retrieval or training runtime

requirements.availability · Optional · one of 99, 99.9, 99.95, 99.99. sets redundancy and operations checks

requirements.p95_latency_ms · Optional · safe integer ≥ 1 or null. adds latency verification

priorities.cost · Optional · safe integer 1–5. steers search and cache

priorities.time_to_market · Optional · safe integer 1–5. weights delivery

priorities.scalability · Optional · safe integer 1–5. weights realtime backend capacity

priorities.performance · Optional · safe integer 1–5. weights realtime throughput

priorities.maintainability · Optional · safe integer 1–5. weights familiarity

priorities.security · Optional · safe integer 1–5. adds resilience at the highest setting

advanced.regulated_use · Optional · one of evaluation, production. Evaluation uses synthetic data. Choose production when real regulated data or live payment processing is required.

advanced.production_review_records · Optional · list of non-empty strings. Enter each review key shown in the results followed by => and your approved review reference. Use a comma between records.

advanced.approved_existing_integrations · Optional · list of non-empty strings. Exact names from Existing Systems approved for reuse. Confirm support status, access, network placement and data handling before adding a name.

advanced.existing_support_records · Optional · list of non-empty strings. Enter each approved system's exact name followed by => and the support review reference for its deployed version.

advanced.realtime_interaction · Optional · one of notifications, concurrent_editing. Choose concurrent editing when multiple people can change the same document at once.

advanced.ledger_model · Optional · one of none, double_entry. Choose double-entry when the app owns balances or posts financial entries.

advanced.offline_access · Optional · one of read_only, queued_writes. Choose queued writes if users must create or edit records without connectivity.

advanced.resumable_uploads · Optional · true/false. Enable when interrupted media uploads must continue without retransmitting the entire file.

advanced.peak_requests_per_second · Optional · safe integer ≥ 0. Peak dynamic HTTP requests per second. Drives load-test and cost guidance.

advanced.rpo_minutes · Optional · safe integer ≥ 0. Maximum tolerable data loss in minutes. Zero requires synchronous durability review.

advanced.rto_minutes · Optional · safe integer ≥ 1. Maximum tolerable recovery time in minutes. Tight targets require failover drills.

advanced.data_sensitivity · Optional · one of public, internal, sensitive. Sensitive records trigger access and storage safeguards.

advanced.portability · Optional · one of flexible, prefer_portable. Prefer a portable search implementation when PostgreSQL is already in the stack.

advanced.search_documents · Optional · safe integer ≥ 0. Expected searchable records. Helps decide when to start with an in-database index.

Research basis: official provider and framework documentation informs regional offerings, static rendering and database search. Android offline guidance informs local storage and sync. AWS recovery guidance distinguishes recovery time and recovery point. See §10 for links.

5. Knowledge base and decision logic

AWS, Azure and Google Cloud profiles bind hosting, database and object storage to the selected regional provider. Existing Microsoft identity influences provider selection only when the cloud choice is unrestricted. VM mode selects a provider VM and self-operated data services. Local mode uses local hosting, build scripts and telemetry. An unknown required region or incompatible air-gap dependency returns an infeasible result.

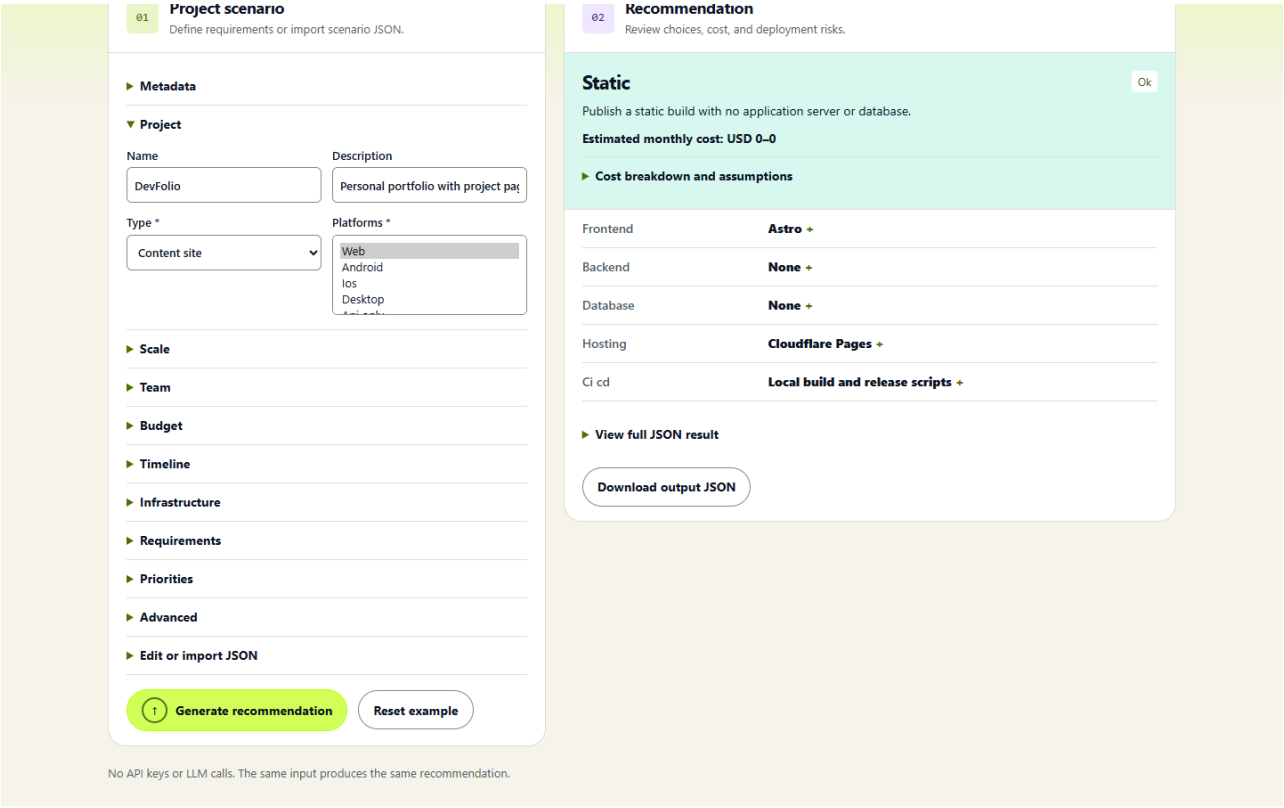
The backend score sums language fit, delivery, realtime throughput and a bounded Microsoft identity ecosystem bonus when applicable. A small or beginner team increases the importance of language fit. A short MVP raises the delivery weight. Performance and scalability priorities raise the realtime term. Ratings are design heuristics. Stable catalog order settles ties. Outputs show winning and runner-up score contributions.

A cost band sums compute, database, storage, identity and applicable supporting workloads. App replica count comes from peak requests and a load-test target of 100 requests per second per replica. Database sizing uses explicit storage assumptions. A minimum above budget blocks the stack. An overlapping band retains the warning and offers a smaller pilot with spending allocations. Taxes, labor, transaction fees, SMS and accelerators are excluded from the base band.

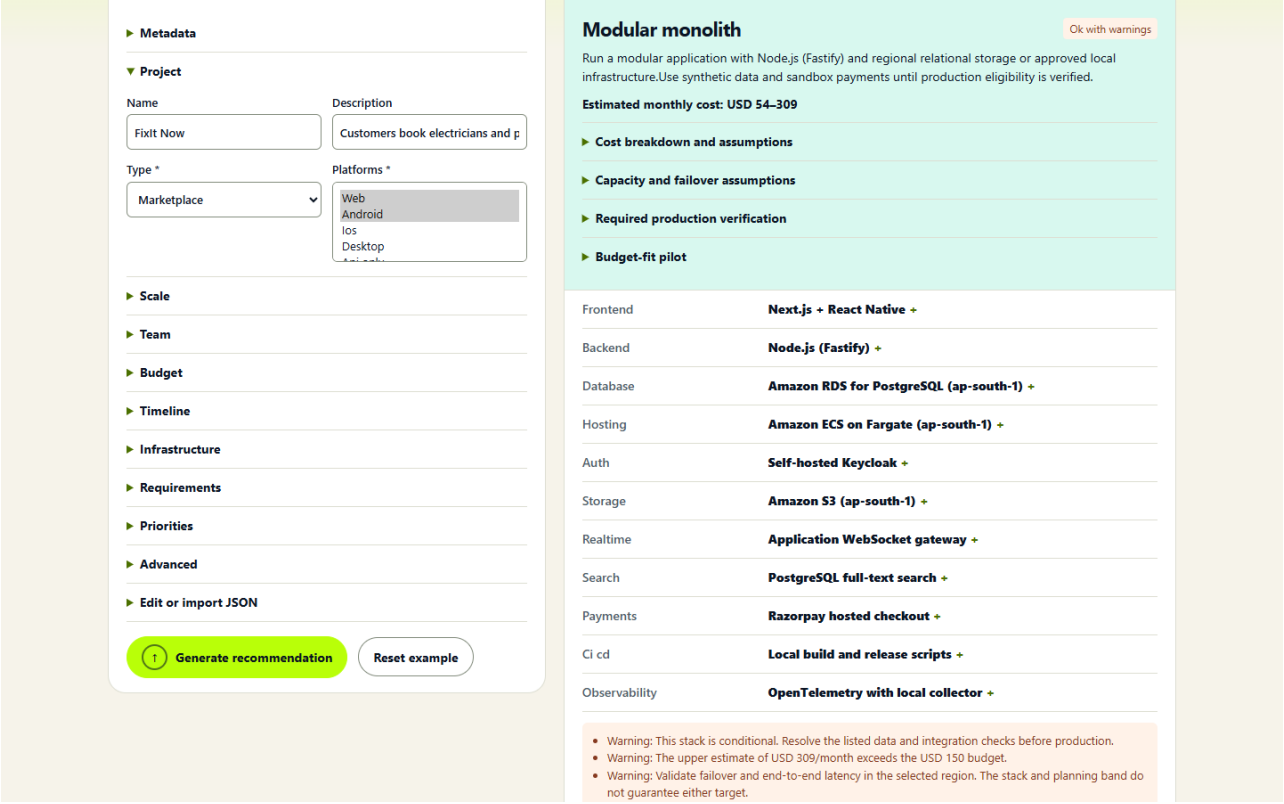
The engine makes no model calls. No API key, network access or feature switch is needed to reproduce the output.

6. Sample recommendations · P1 and P2

P1 / Astro / Cloudflare Pages / USD 0-0 per month.

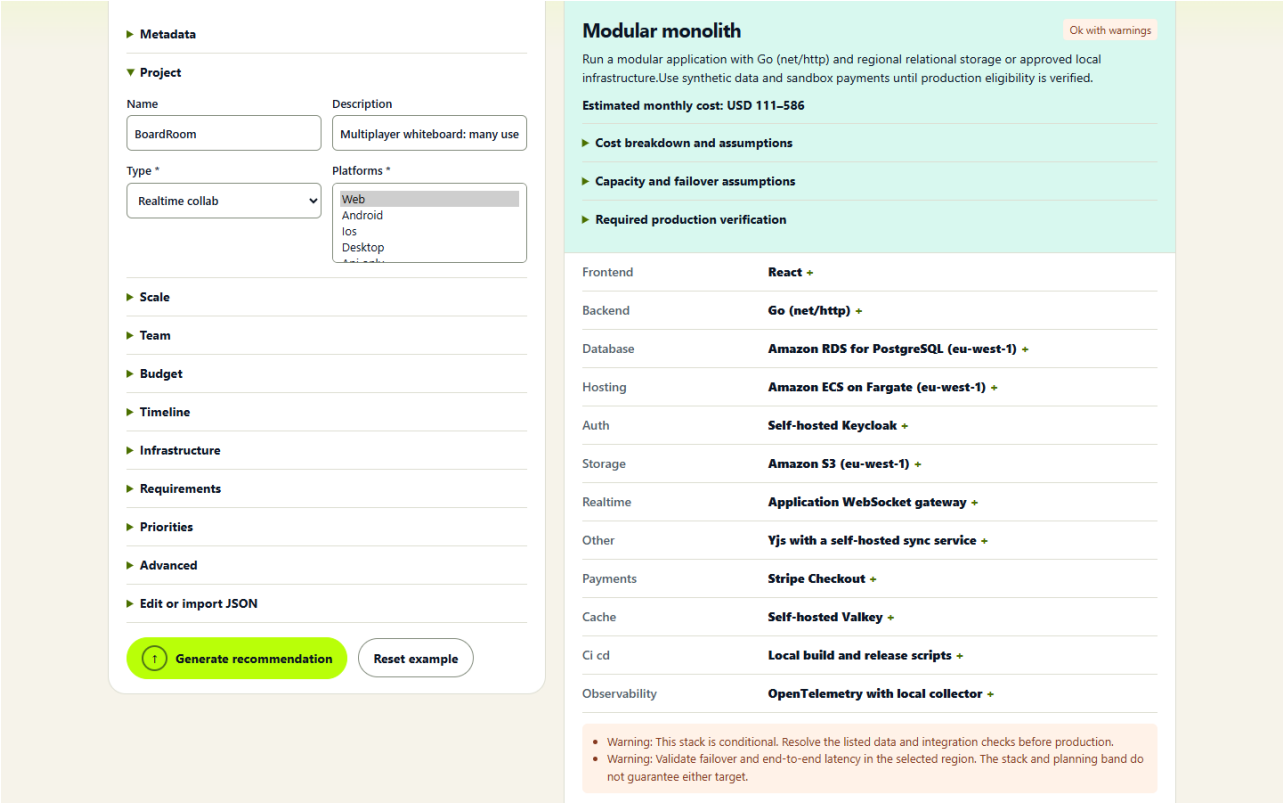


P2 / Next.js + React Native / Amazon ECS on Fargate (ap-south-1) / USD 54-309 per month.

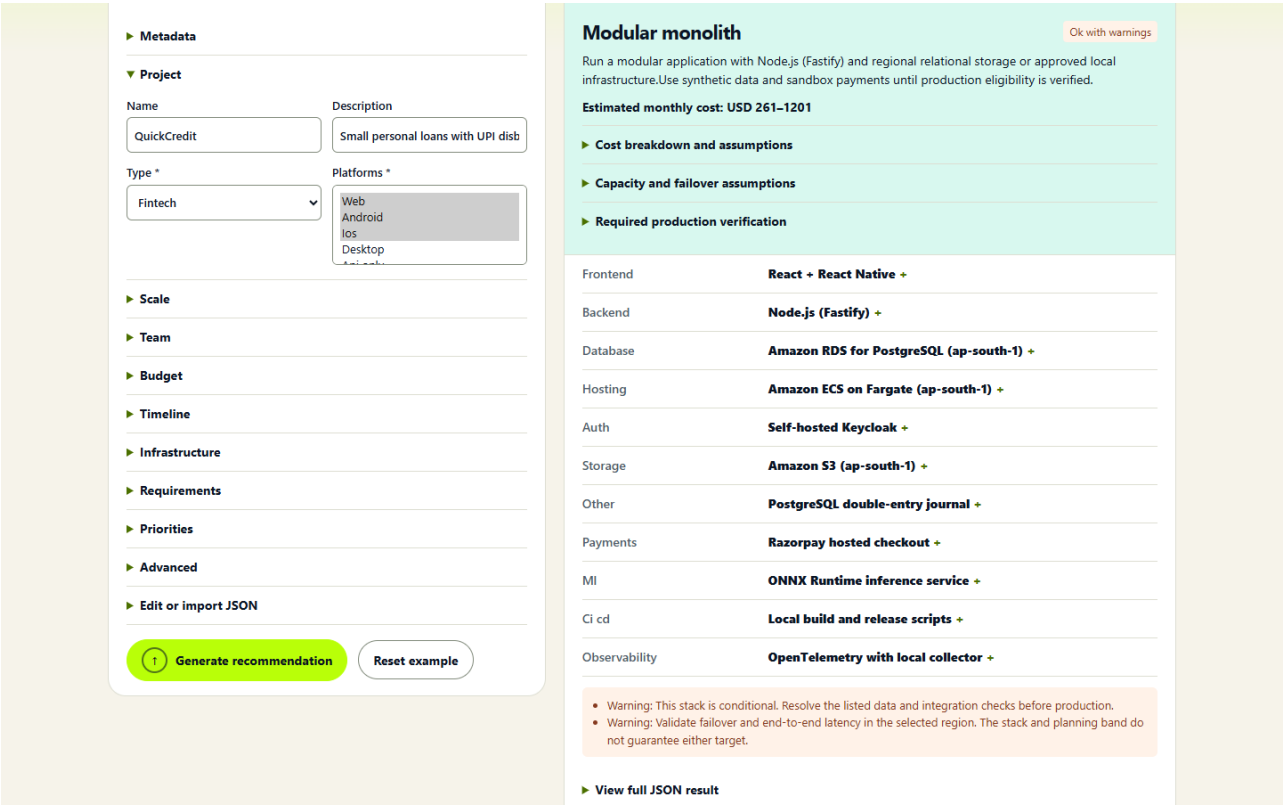


6. Sample recommendations · P3 and P4

P3 / React / Amazon ECS on Fargate (eu-west-1) / USD 111-586 per month.

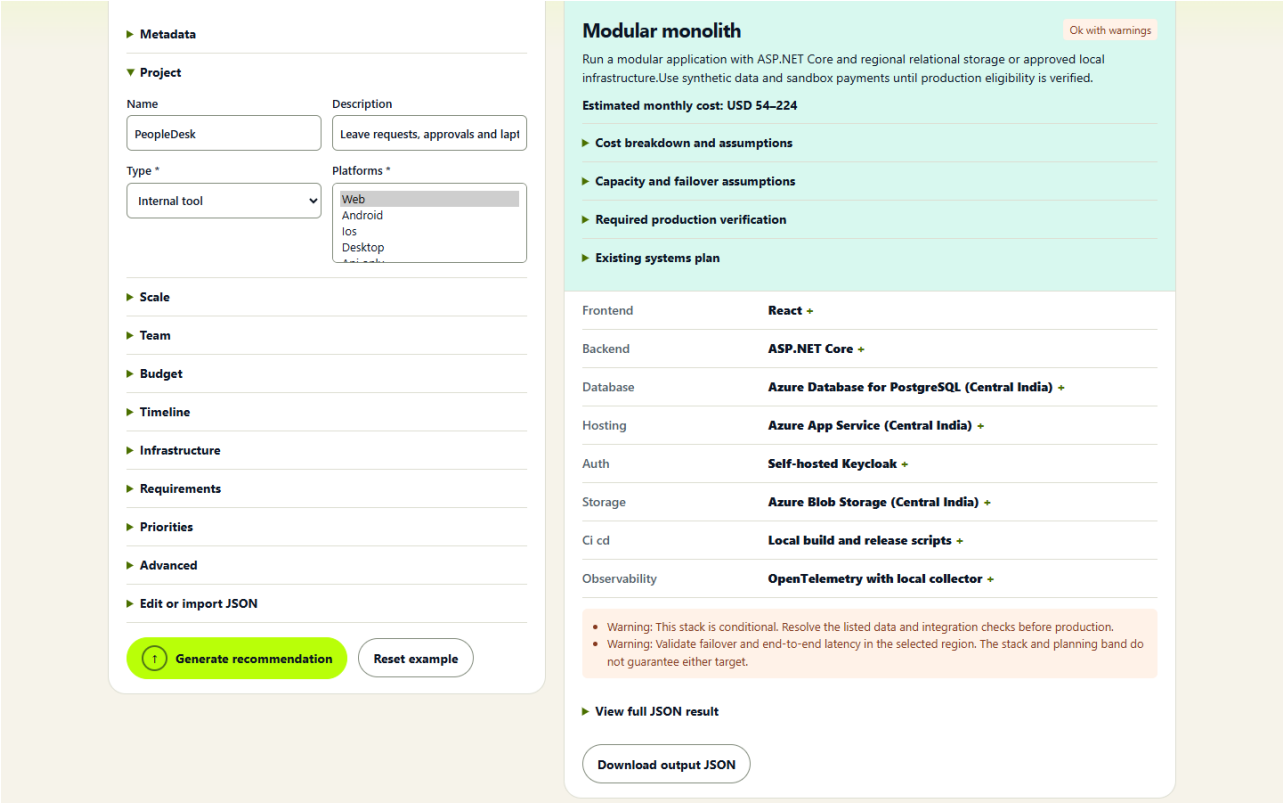


P4 / React + React Native / Amazon ECS on Fargate (ap-south-1) / USD 261-1201 per month.

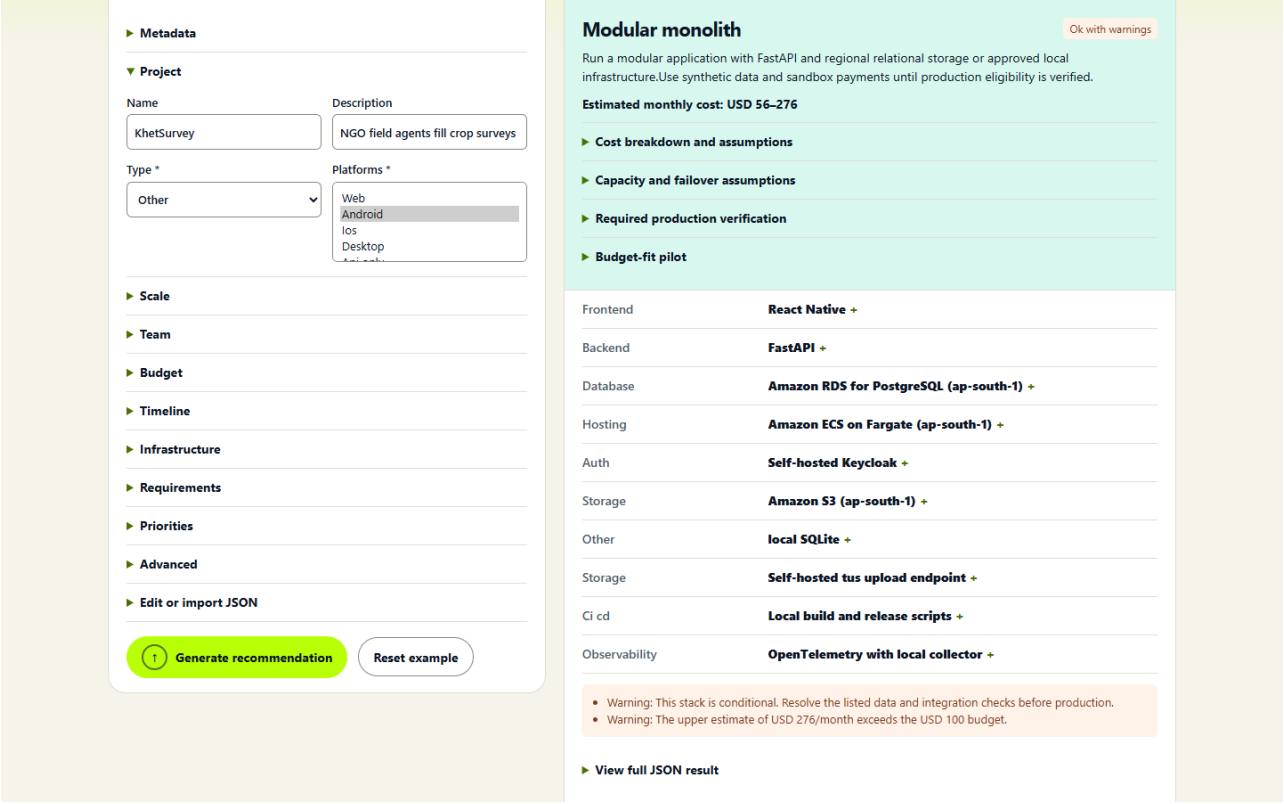


6. Sample recommendations · P5 and P6

P5 / React / Azure App Service (Central India) / USD 54-224 per month.



P6 / React Native / Amazon ECS on Fargate (ap-south-1) / USD 56-276 per month.



6. Component rationale · P2

Frontend · Next.js + React Native. Covers the requested web, android clients.

Backend · Node.js (Fastify). Scored 31.2 (language fit 15.0, delivery 15.0, realtime 1.2). Matches declared skills: typescript, javascript.

Database · Amazon RDS for PostgreSQL (ap-south-1). Relational storage provides transactions and constraints for a balanced workload.

Hosting · Amazon ECS on Fargate (ap-south-1). Run the workload in ap-south-1 using the selected deployment model.

Auth · Self-hosted Keycloak. Keep identity data on the chosen infrastructure. Add an SMS adapter and verify carrier terms.

Storage · Amazon S3 (ap-south-1). Keep uploads and backups inside the chosen deployment boundary.

Realtime · Application WebSocket gateway. Host the connection and fan-out path with the application. Use a continuously available service and verify connection duration limits.

Search · PostgreSQL full-text search. Reuse the transactional database and avoid operating a second search service.

Payments · Razorpay hosted checkout. Use hosted checkout and keep card entry outside the application.

Ci_Cd · Local build and release scripts. Keep builds portable and publish only approved artifacts.

Observability · OpenTelemetry with local collector. Collect telemetry without requiring a third-party endpoint.

P2 pilot · USD 130.77 per month, with USD 19.23 unallocated. Start with one rollout cohort. Preserve realtime and payment flows. Gate new enrollment on observed session and transaction costs.

P6 pilot · USD 91.20 per month, with USD 8.80 unallocated. Start with one field cohort. Preserve offline capture and queued uploads. Defer onboarding when sync backlog or metered usage exceeds the pilot envelope.

6. Component rationale · P5

Frontend · React. Covers the requested web clients.

Backend · ASP.NET Core. Scored 43.0 (language fit 27.0, delivery 10.0, identity ecosystem 6.0). Matches declared skills: csharp.

Database · Azure Database for PostgreSQL (Central India). Relational storage provides transactions and constraints for a read heavy workload.

Hosting · Azure App Service (Central India). Run the workload in Central India using the selected deployment model.

Auth · Self-hosted Keycloak. Keep identity data on the chosen infrastructure. Broker the organization’s existing IdP over OIDC or SAML.

Storage · Azure Blob Storage (Central India). Keep uploads and backups inside the chosen deployment boundary.

Ci_Cd · Local build and release scripts. Keep builds portable and publish only approved artifacts.

Observability · OpenTelemetry with local collector. Collect telemetry without requiring a third-party endpoint.

7. Edge-case handling

Category	Detection	Engine response	Capture
Missing fields	Required schema keys absent	Return invalid_input with field errors. List applied defaults in assumptions.	E-missing
Invalid values	Schema type, enum or range fails	Return field errors. The capture uses a string where a Boolean is required.	E-type
Contradictions	Budget below the modeled floor or incompatible hard constraints	Return infeasible with a critical warning. Tests also cover solo availability and short multiplatform MVPs.	E-budget
Over-engineering	Content-only web site without dynamic requirements	Keep the site static even at high forecast traffic. Verify free-tier limits before launch.	E-static
Regulatory constraints	Residency and compliance flags or a production request	Filter regions, list affected services and block unreviewed production. Evaluation uses synthetic data.	E-regulated
Existing systems	Declared system with matching reuse and support review records	Keep existing records behind a read-only adapter. The capture uses a demonstration review reference.	E-existing
Restricted deployment	On-premises or air-gapped boundary	Serve and build locally. Reject dependencies that require external access.	E-airgap
Skill mismatch	Declared skills differ from the default stack	Weight language fit. The C# mutation selects ASP.NET Core and explains its score.	E-skills
Deprecated technology	Known retirement matches the lifecycle catalog	Warn and propose migration. Block explicit reuse of the retired service.	E-retired

Appendix A contains the corresponding unedited screenshots from the running browser engine.

8. Validation

The Node suite checks deterministic output, cost totals, provider changes, feasibility, platform coverage, lifecycle and review gating. Chrome imported all six public inputs and nine edge cases, generated each recommendation, and compared each result with Node. The capture script verified the browser export filename and the generated JSON payload. The screenshots are raw browser captures.

Scenario checks confirm that P1 stays static, P3 selects an EU region for GDPR evaluation, P4 retains India residency, P5 favors the declared C# skills, and P6 retains local capture and resumable upload support. These are behavior checks against the provided requirements and documented design patterns.

9. Limitations and future work

The catalog and ratings are curated rules. Throughput and cost require regional quotes and workload benchmarks. Production review references are supplied by users and require independent review. Availability and latency are targets to test. Expand lifecycle coverage, calibrate estimates and model tenant isolation and legacy migrations in more detail.

10. References and third-party tools

Sources include the organizer-supplied StackSense Challenge problem statement and public dataset. Runtime uses Node.js and browser APIs with zero third-party runtime libraries or external API calls. Python and ReportLab generate the PDF. Playwright and Chrome capture the browser evidence.

Supplied public scenarios

Python

Challenge listing

aws.amazon.com/fargate/pricing/

aws.amazon.com/pricing/

aws.amazon.com/rds/postgresql/pricing/

aws.amazon.com/s3/pricing/

azure.microsoft.com/pricing/

azure.microsoft.com/pricing/details/app-service/linux/

azure.microsoft.com/pricing/details/postgresql/flexible-server/

azure.microsoft.com/pricing/details/storage/blobs/

cloud.google.com/run/docs/locations

cloud.google.com/run/pricing

cloud.google.com/sql/pricing

cloud.google.com/storage/pricing

developer.android.com/topic/architecture/data-layer/offline-first

developer.mozilla.org/en-US/docs/Web/API

developer.mozilla.org/en-US/docs/Web/API/IndexedDB_API

developers.cloudflare.com/pages/platform/limits/

docs.astro.build/

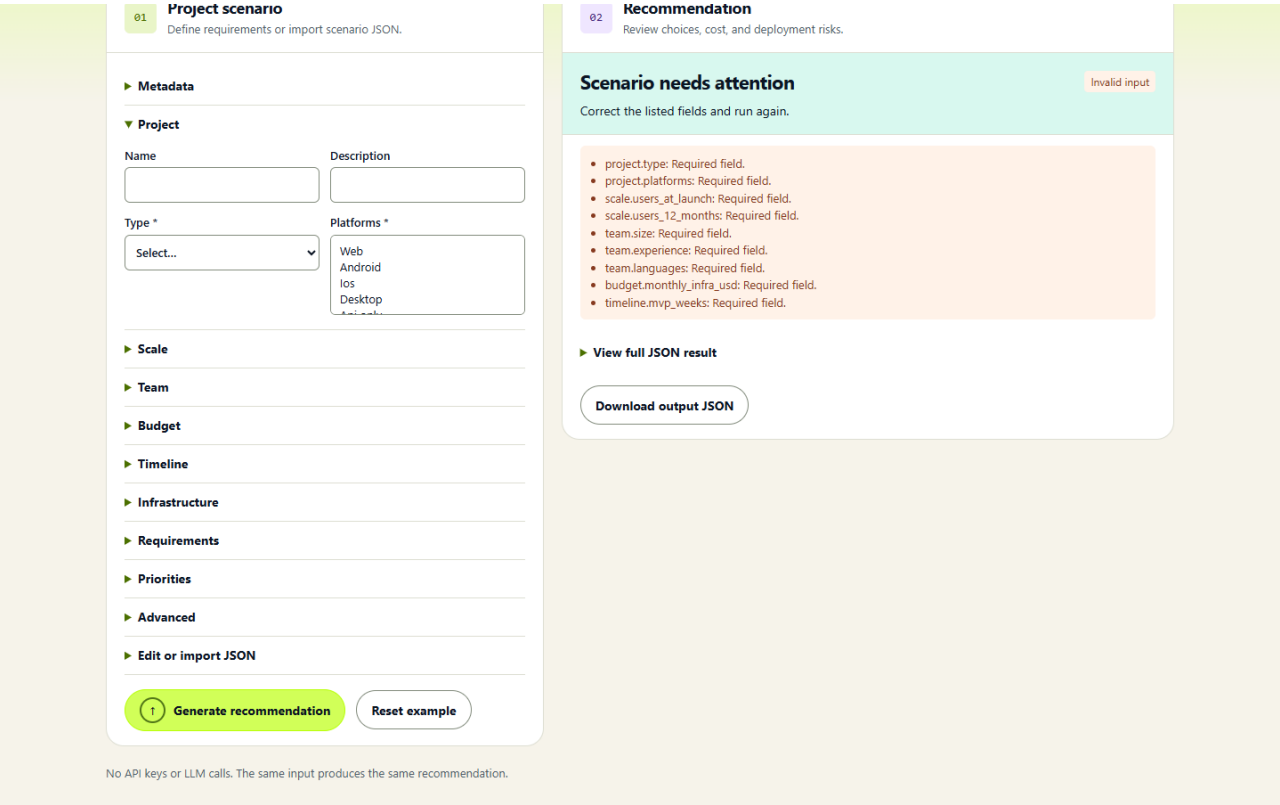
docs.astro.build/en/concepts/islands/
docs.aws.amazon.com/wellarchitected/latest/reliability-pillar/plan-for-disaster-recovery-dr.html
docs.yjs.dev/
fastapi.tiangolo.com/
fastify.dev/docs/latest/
learn.microsoft.com/aspnet/core/
learn.microsoft.com/en-us/azure/backup/backup-azure-database-postgresql
learn.microsoft.com/en-us/lifecycle/products/azure-database-for-mysql-single-server
learn.microsoft.com/en-us/sql/sql-server/end-of-support/sql-server-end-of-support-overview
learn.microsoft.com/lifecycle/announcements/sql-server-2012-windows-server-2012-r2-end-of-support
nginx.org/en/docs/
nodejs.org/
opentelemetry.io/docs/
opentelemetry.io/docs/collector/
pagefind.app/docs/
pkg.go.dev/net/http
playwright.dev/
react.dev/
spring.io/projects/spring-boot
spring.io/projects/spring-boot/
tus.io/protocols/resumable-upload
valkey.io/topics/
www.cloudflare.com/developer-platform/products/pages/
www.google.com/chrome/
www.hhs.gov/hipaa/for-professionals/special-topics/health-information-technology/cloud-computing/index.html
www.keycloak.org/docs/latest/server_admin/
www.keycloak.org/documentation
www.meilisearch.com/docs
www.pcisecuritystandards.org/faqs/1092/
www.postgresql.org/docs/
www.postgresql.org/docs/current/
www.postgresql.org/docs/current/textsearch.html
www.postgresql.org/docs/current/tutorial-transactions.html
www.python.org/
www.reportlab.com/
www.rfc-editor.org/rfc/rfc6455

11. AI assistance declaration

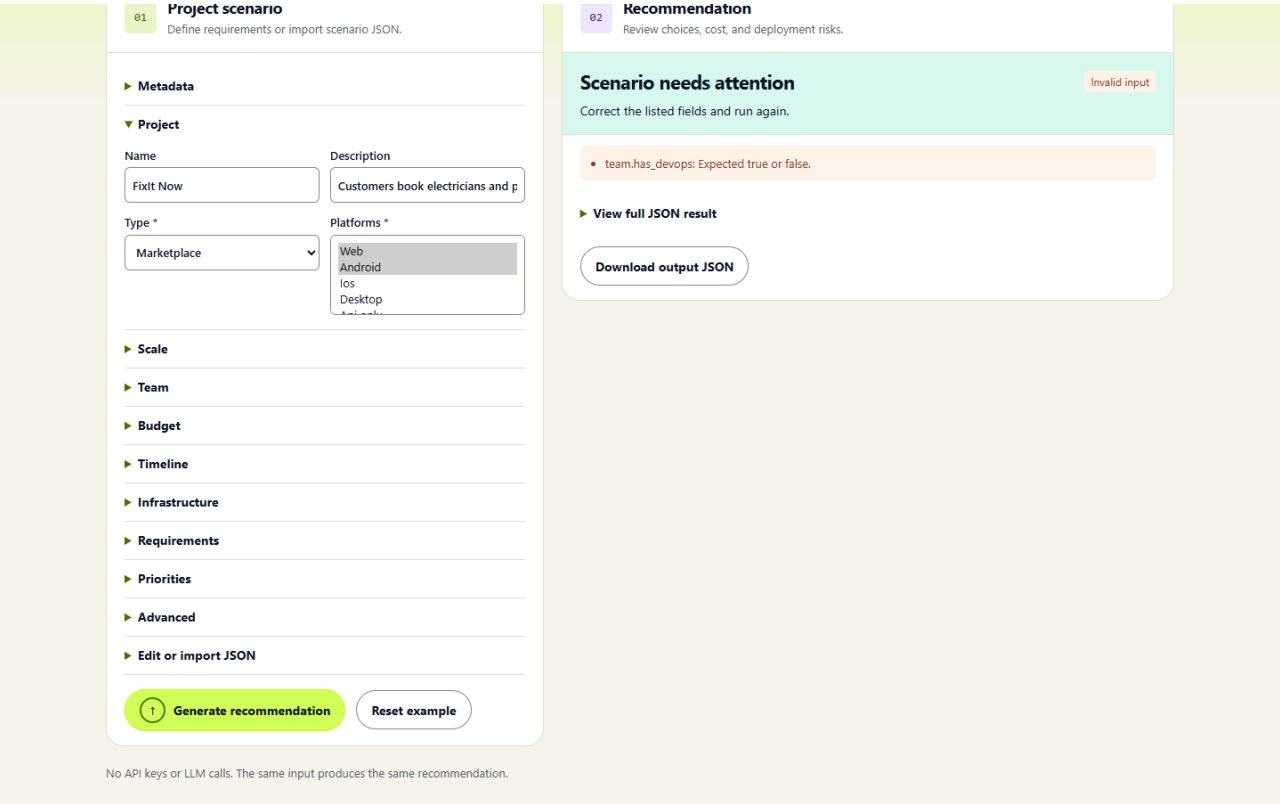
OpenAI Codex assisted with implementation, interface changes, test design, report drafting and evidence capture. The runtime uses deterministic code and makes no model calls.

Appendix A. Edge-case screenshots

E-missing · Missing fields

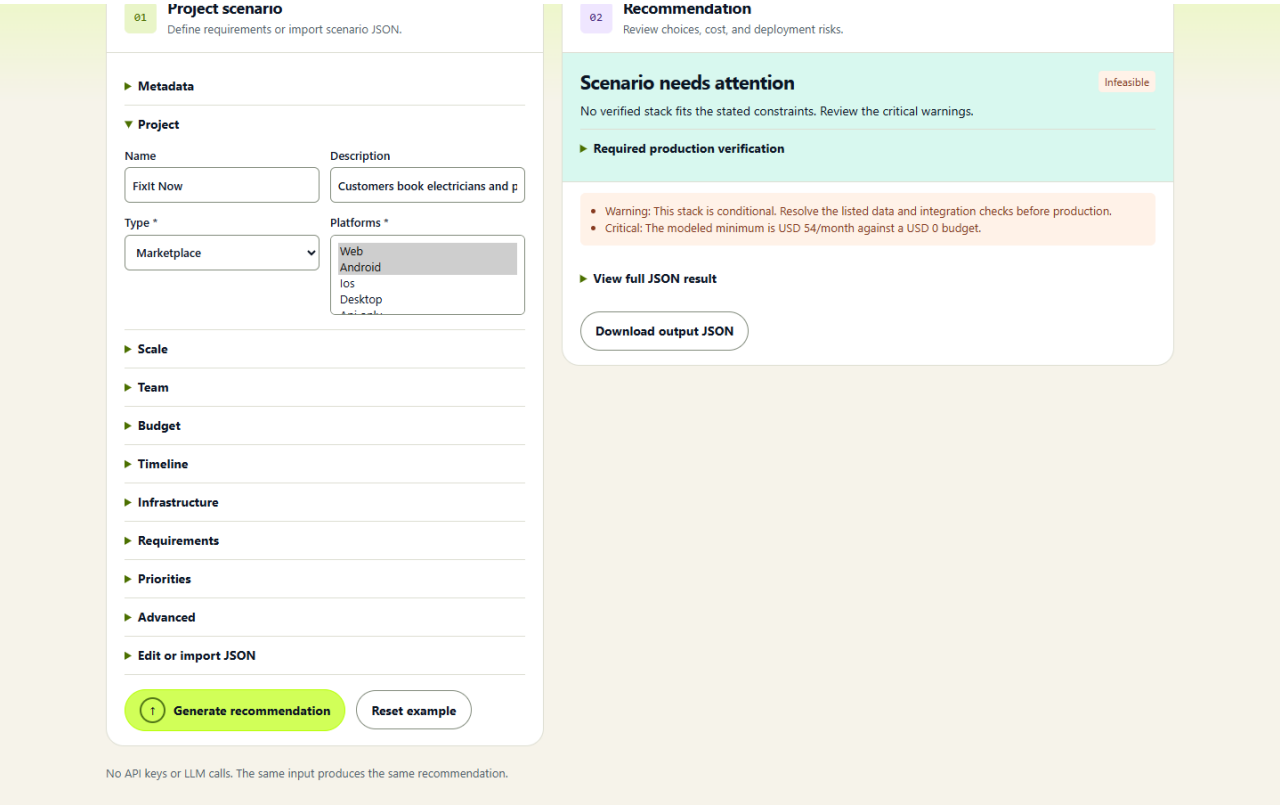


E-type · Invalid values

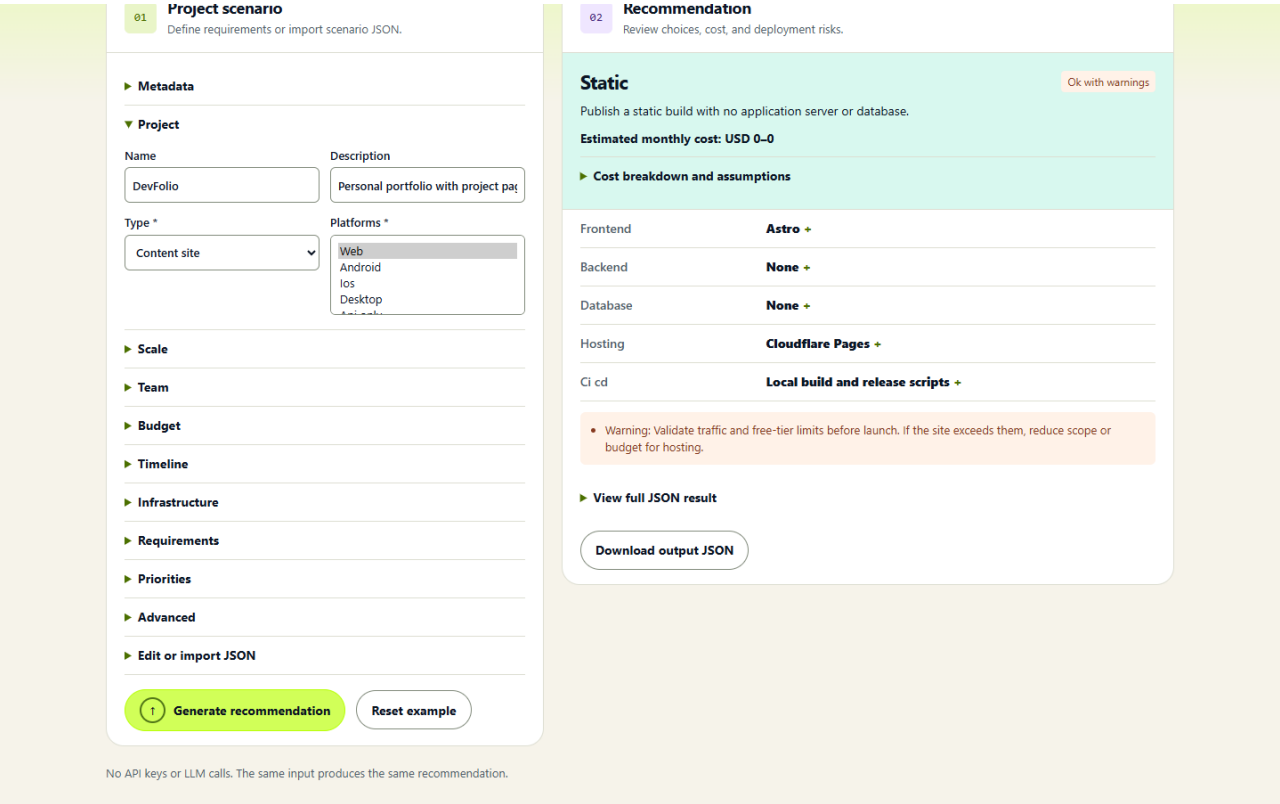


Appendix A. Edge-case screenshots

E-budget · Contradictions

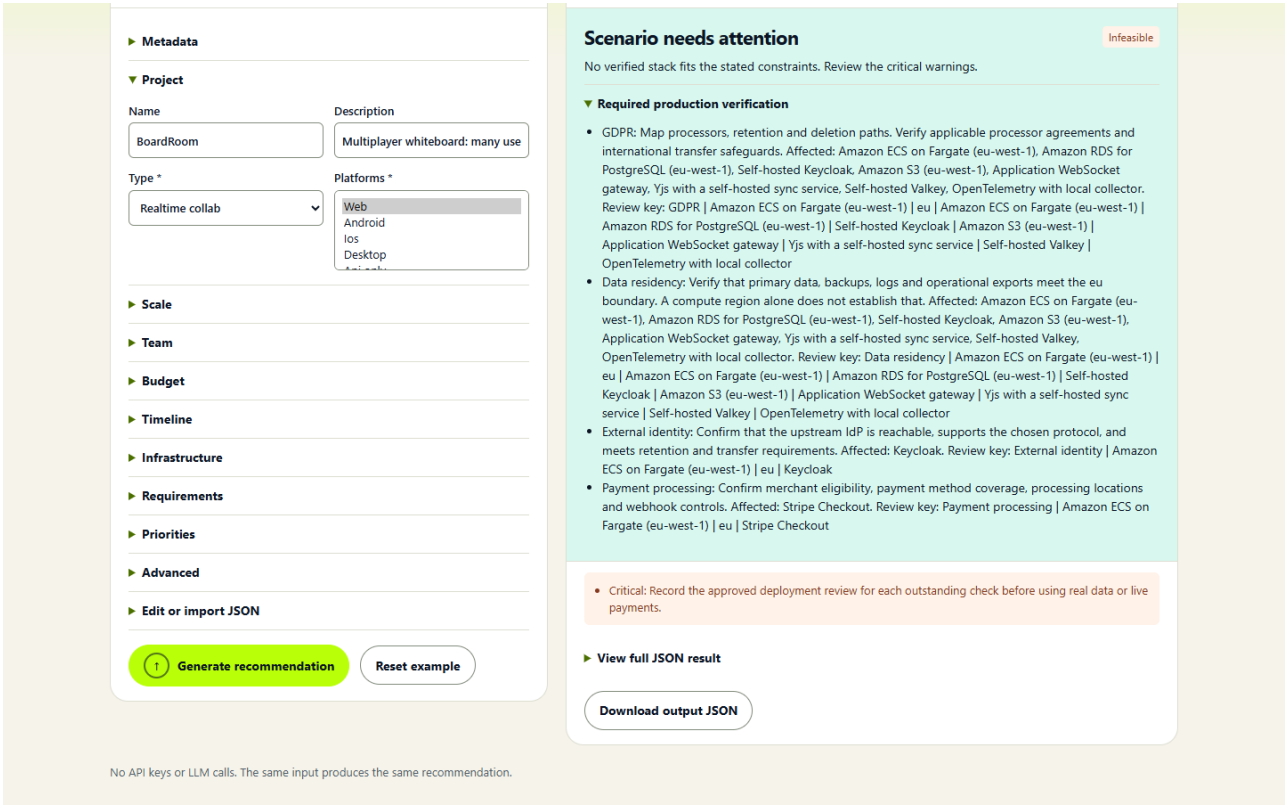


E-static · Over-engineering

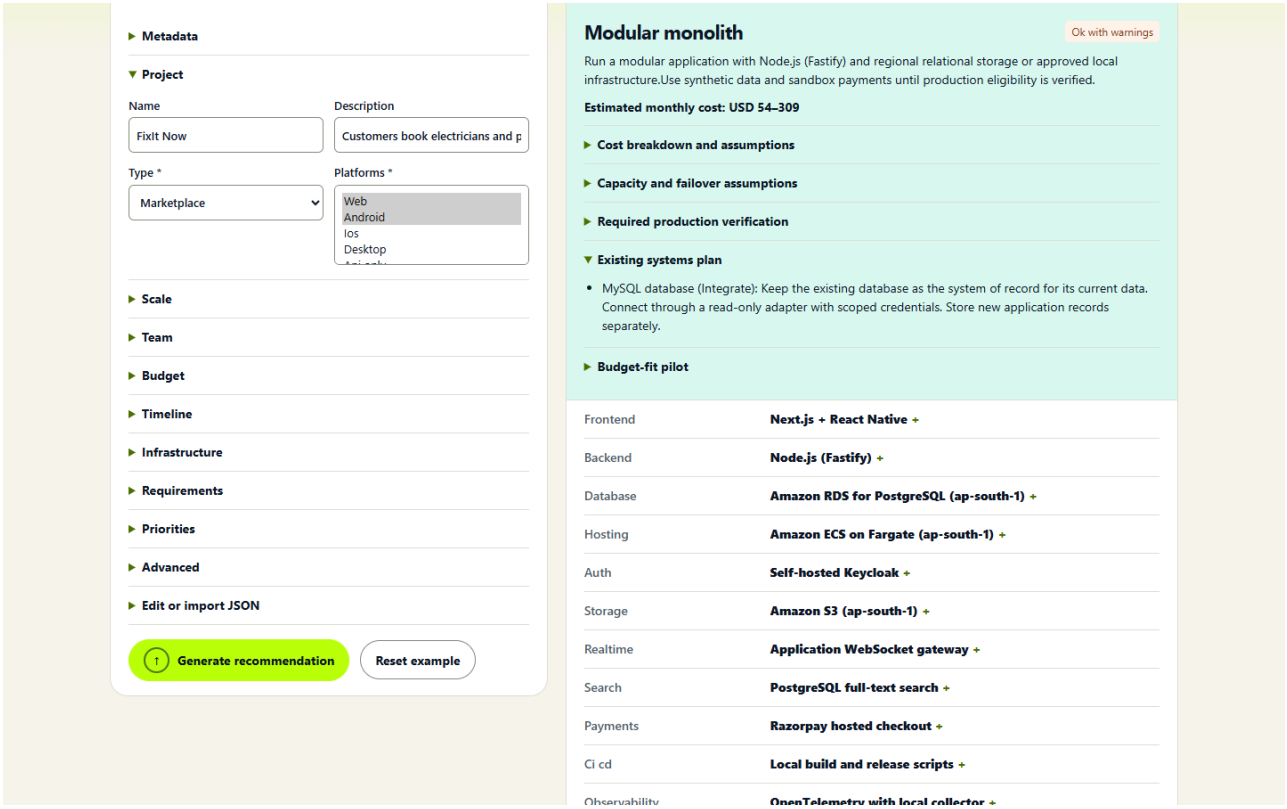


Appendix A. Edge-case screenshots

E-regulated · Regulatory constraints

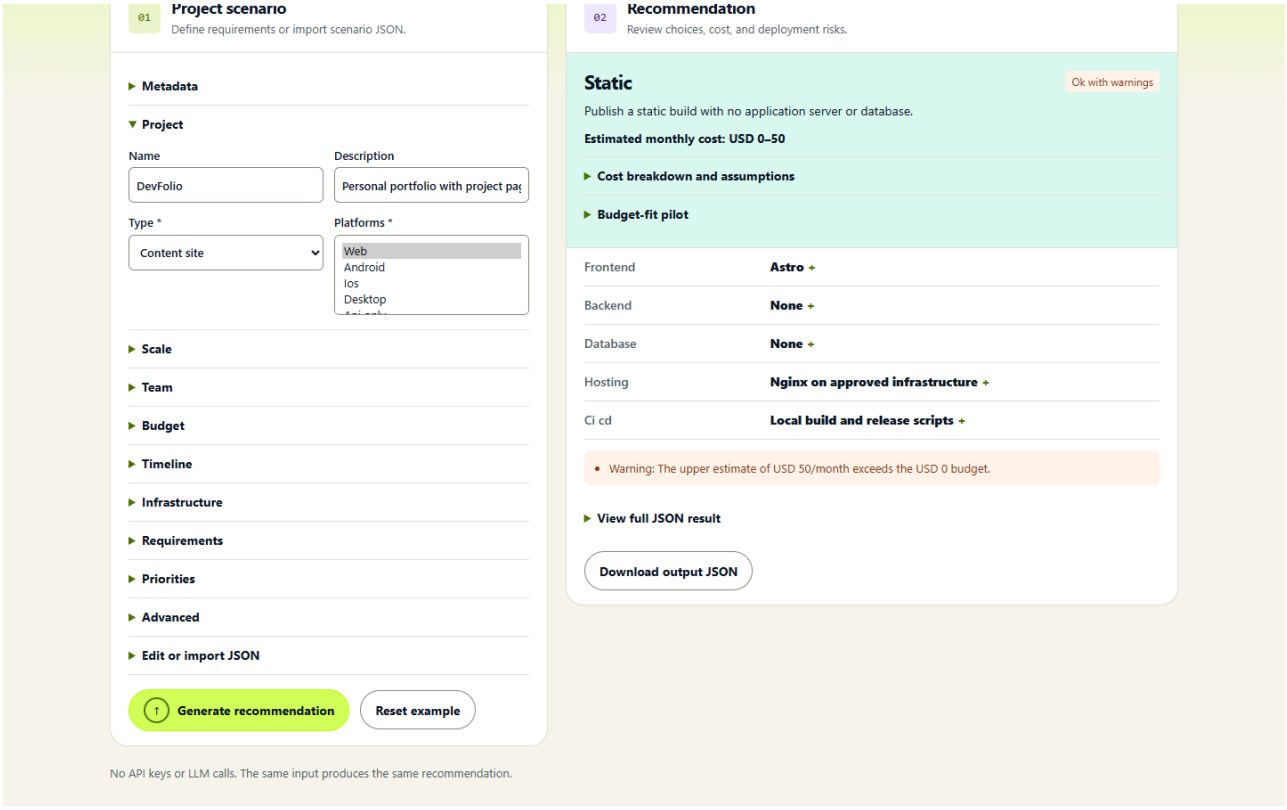


E-existing · Existing systems

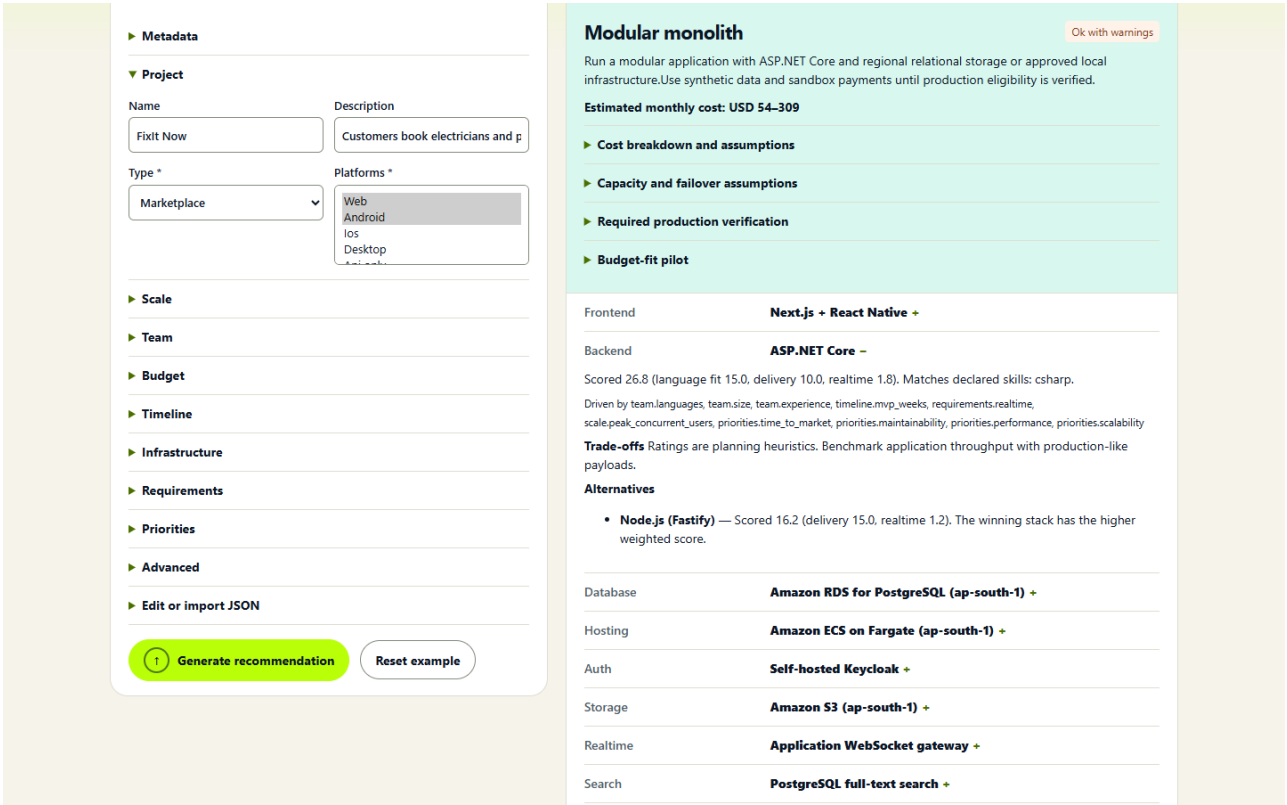


Appendix A. Edge-case screenshots

E-airgap · Restricted deployment



E-skills · Skill mismatch



Appendix A. Edge-case screenshots

E-retired · Deprecated technology

